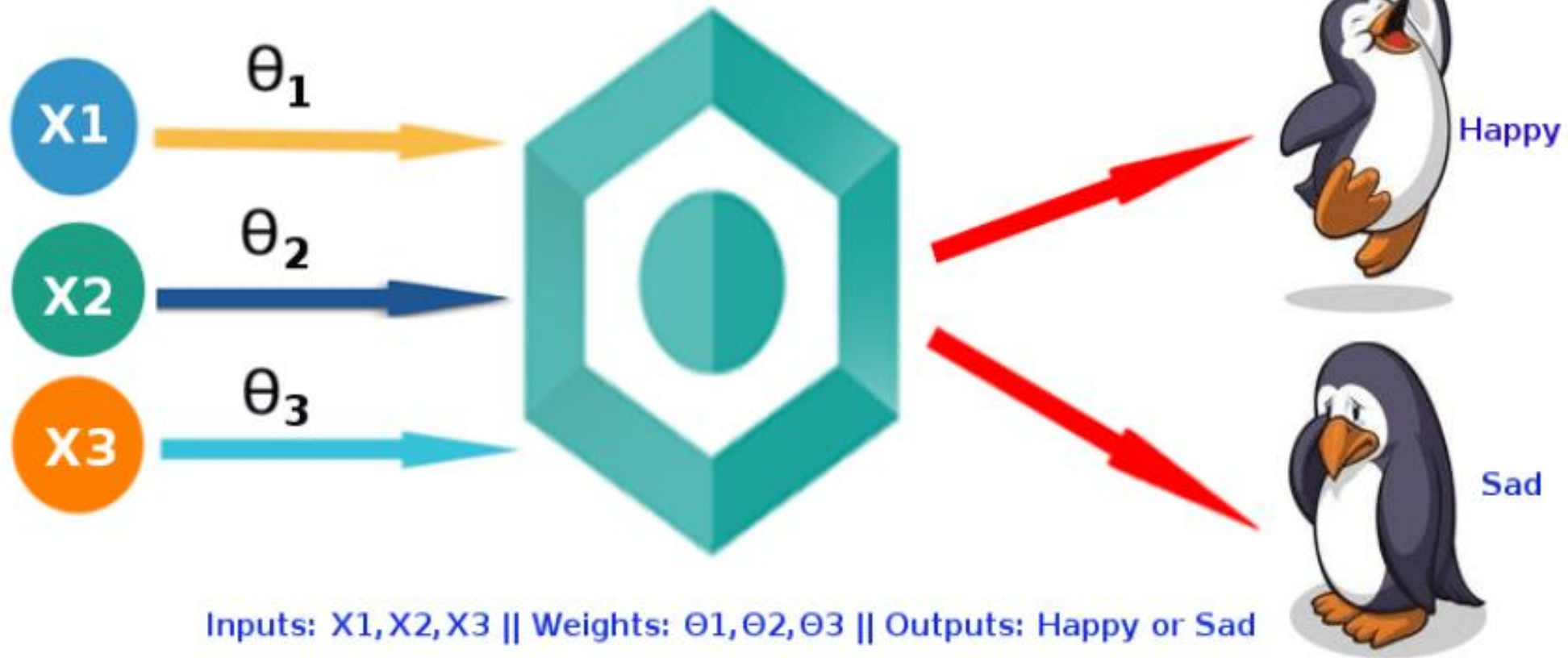


Logistic Regression

Introduction of Logistic Regression

Logistic Regression Model



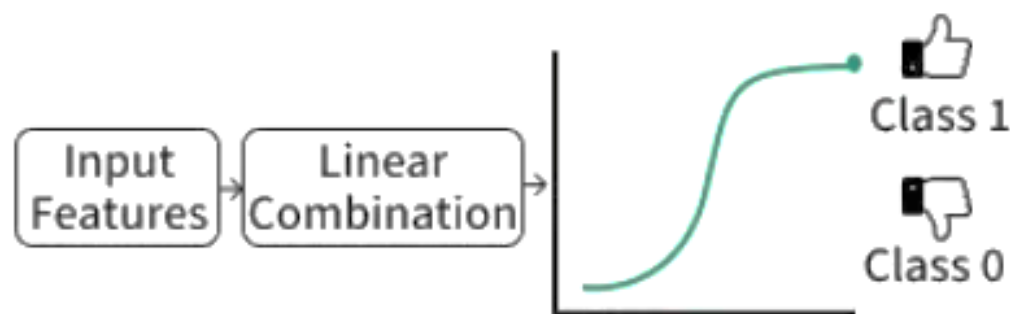
Logistic Regression was used in the biological sciences in early twentieth century. It was then used in many social science applications. Logistic Regression is used when the dependent variable(target) is categorical.

For example,

- To predict whether an email is spam (1) or (0)
- Whether the tumor is malignant (1) or not (0)

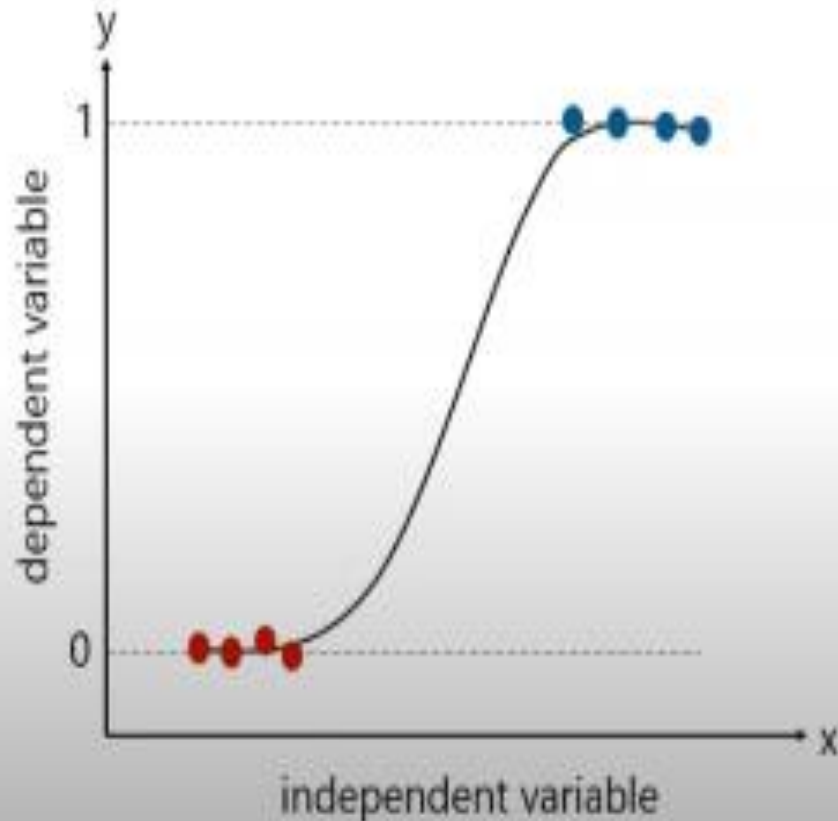
What is Logistic Regression?

- Predicts the probability of a binary outcome (Yes/No, 0/1)
- Uses the sigmoid function to map inputs to probabilities (0 to 1)
- Ideal for classification tasks



What is Logistic Regression

Logistic Regression is a method used to predict a dependent variable, given a set of independent variables, such that the dependent variable is categorical.



- *Dependent variable (Y):*
The response binary variable holding values like 0 or 1, Yes or No, A, B or C
- *Independent variable (X):*
The predictor variable used to predict the response variable.

The following equation is used to represent a linear regression model:

$$\log \left(\frac{Y}{1 - Y} \right) = C + B_1X_1 + B_2X_2 + \dots$$

Need of Logistic Regression

Consider a scenario where we need to classify whether an email is spam or not. If we use linear regression for this problem, there is a need for setting up a threshold based on which classification can be done. Say if the actual class is malignant, predicted continuous value 0.4 and the threshold value is 0.5, the data point will be classified as not malignant which can lead to serious consequence in real time.

From this example, it can be inferred that linear regression is not suitable for classification problem. Linear regression is unbounded, and this brings logistic regression into picture. Their value strictly ranges from 0 to 1.

Understanding Sigmoid Function

1. The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.
2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the **sigmoid curve or logistic curve**. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.
3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.
 - If the sigmoid output is same or above the threshold, the input is classified as **Class 1**.
 - If it is below the threshold, the input is classified as **Class 0**.

Model

Output = 0 or 1

Hypothesis $\Rightarrow Z = WX + B$

$h\Theta(x) = \text{sigmoid}(Z)$

Sigmoid Function

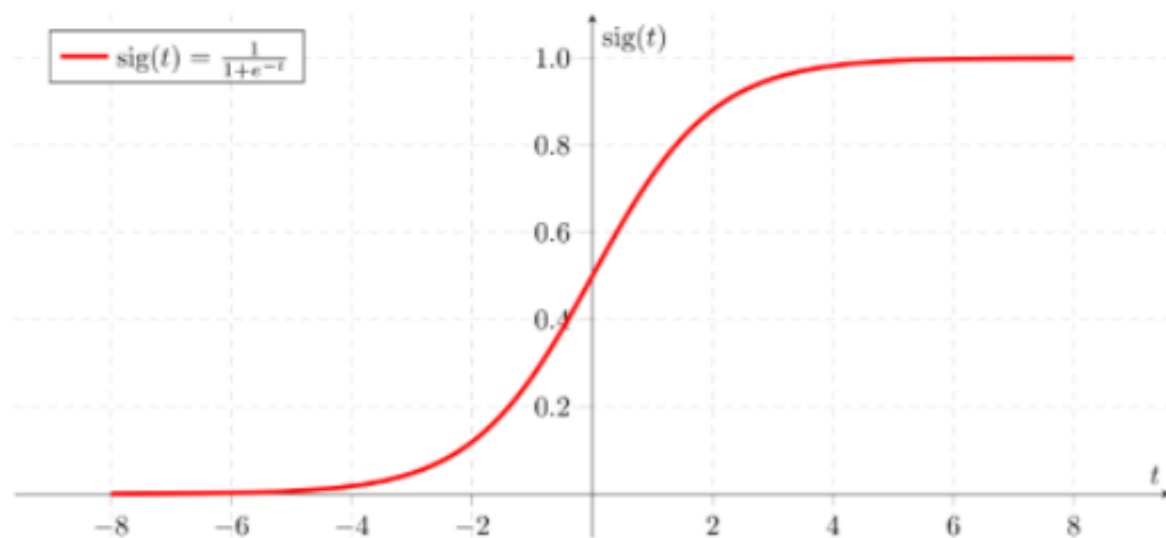


Figure 2: Sigmoid Activation Function

If 'Z' goes to infinity, Y(predicted) will become 1 and if 'Z' goes to negative infinity, Y(predicted) will become 0.

Logistic Regression Types

- 1. Binomial Logistic Regression:** This type is used when the dependent variable has only two possible categories. Examples include Yes/No, Pass/Fail or 0/1. It is the most common form of logistic regression and is used for binary classification problems.
- 2. Multinomial Logistic Regression:** This is used when the dependent variable has three or more possible categories that are not ordered. For example, classifying animals into categories like "cat," "dog" or "sheep." It extends the binary logistic regression to handle multiple classes.
- 3. Ordinal Logistic Regression:** This type applies when the dependent variable has three or more categories with a natural order or ranking. Examples include ratings like "low," "medium" and "high." It takes the order of the categories into account when modeling.

Types of Logistic Regression

Binomial

Two classes (0 or 1)



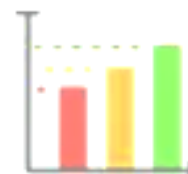
Multinomial

More than two unordered classes (cat, dog, sheep)



Ordinal

More than two ordered classes (low, medium, high)



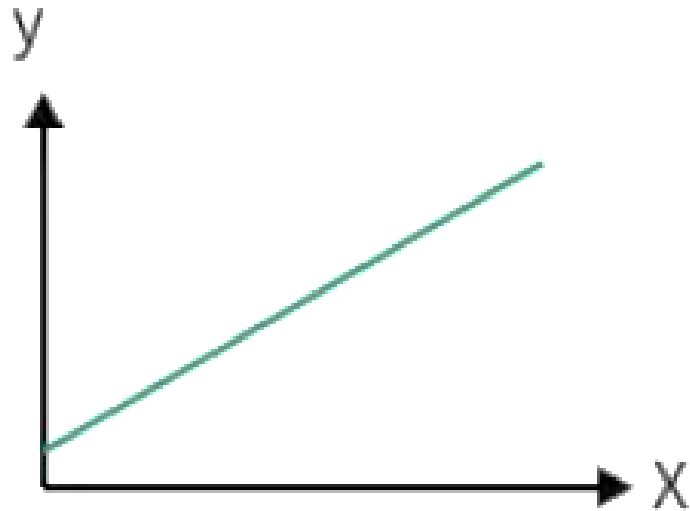
Difference between Linear and Logistic Regression

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic regression is used to predict the categorical dependent variable using a given set of independent variables.
It is used for solving regression problem.	It is used for solving classification problems.
In this we predict the value of continuous variables	In this we predict values of categorical variables

In this we find best fit line.	In this we find S-Curve.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for Estimation of accuracy.
The output must be continuous value, such as price, age etc.	Output must be categorical value such as 0 or 1, Yes or no etc.
It required linear relationship between dependent and independent variables.	It not required linear relationship.
There may be collinearity between the independent variables.	There should be little to no collinearity between independent variables.

Linear Regression

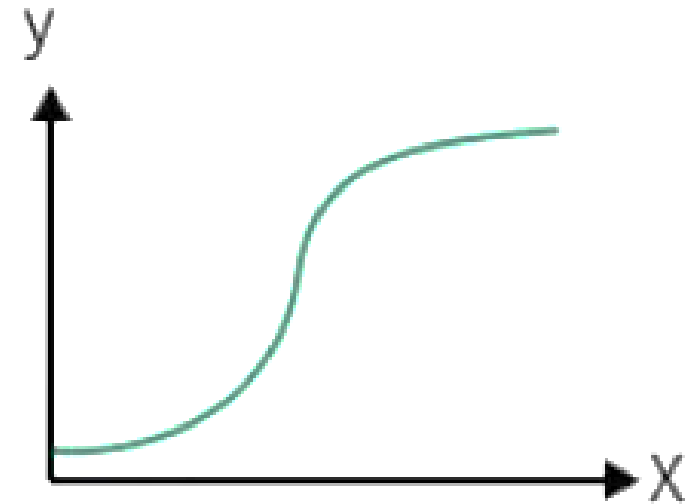
- Predicts continuous values
- Uses best-fit line
- Solves regression problems



vs

Logistic Regression

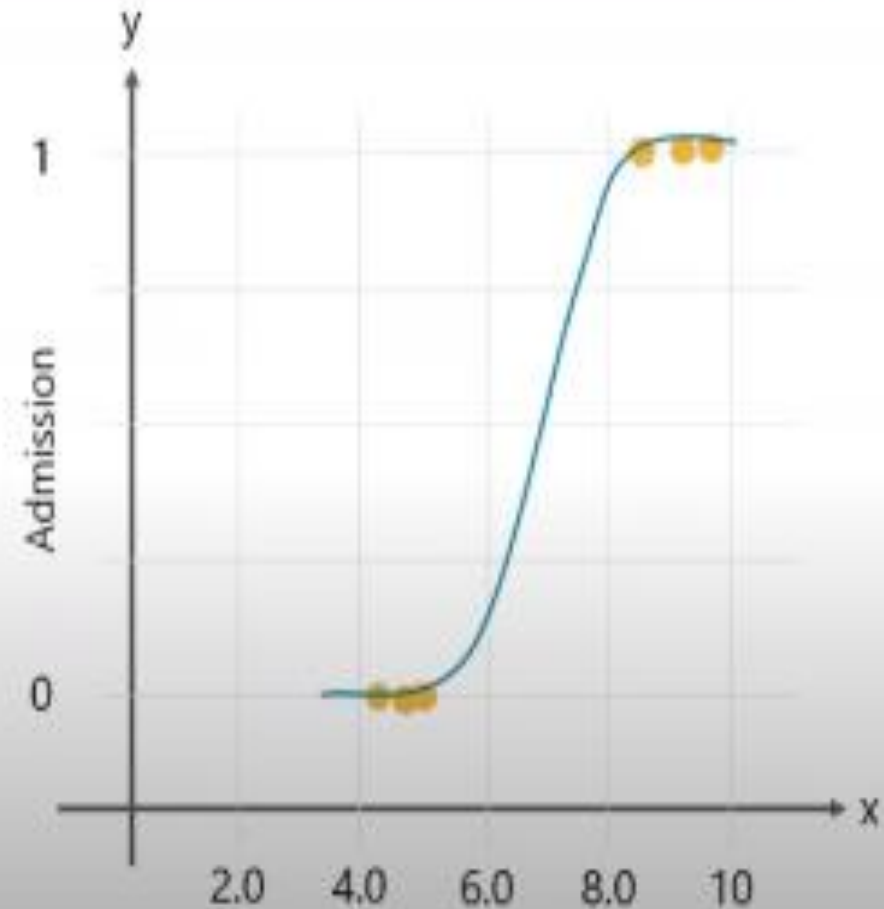
- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems



Use Case of Logistic Regression

To predict if a student will get admitted to a school based on his CGPA.

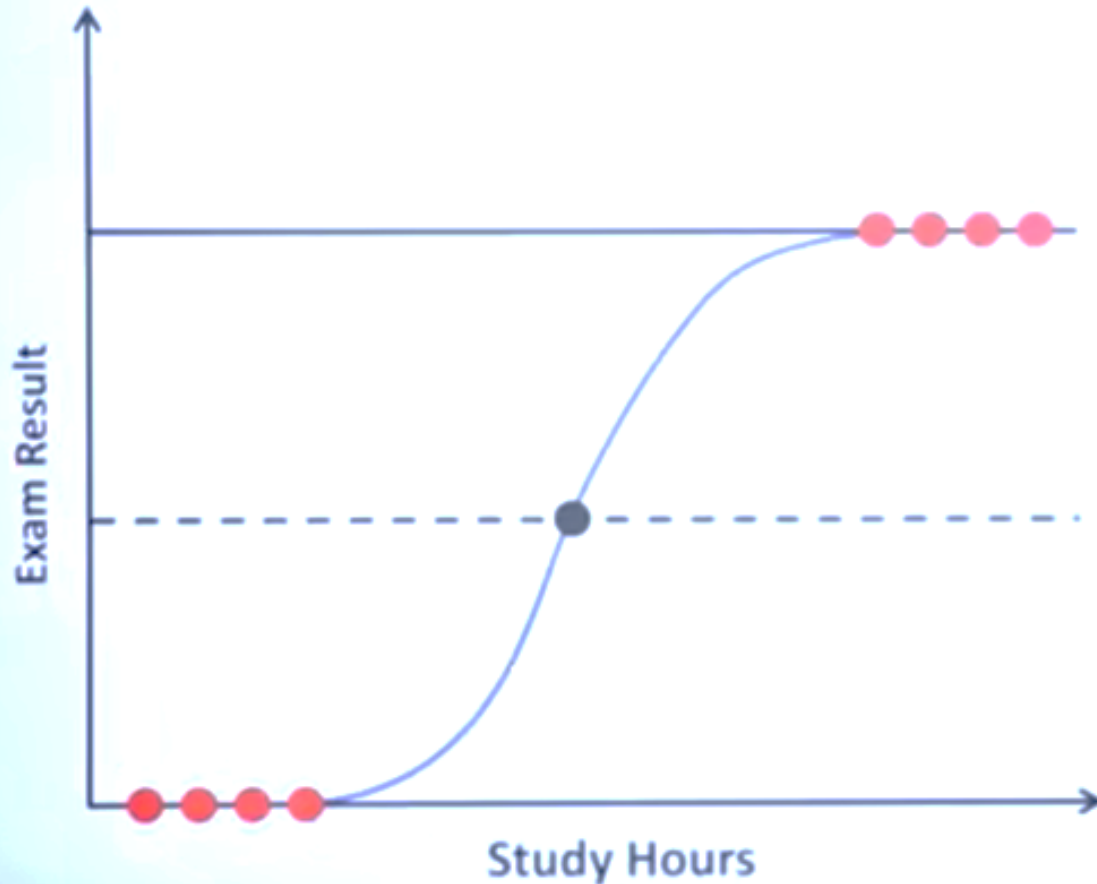
Admission	CGPA
0	4.2
0	5.1
0	5.5
1	8.2
1	9.0
1	9.1



Example (1) of Logistic Regression

$$y = \frac{1}{1 + e^{-(a_0 + a_1 x)}}$$

$$a_0 = -1.5, a_1 = 0.6$$



X (study hours) = 5 hrs / 1.5 hrs

Y (exam result) = ??

Example (2) of Logistic Regression

- The dataset of pass or fail in an exam of 5 students is given in the table.
 - Use logistic regression as classifier to answer the following questions.
1. Calculate the probability of pass for the student who studied 33 hours.
 2. At least how many hours student should study that makes he will pass the course with probability of more than 95%.

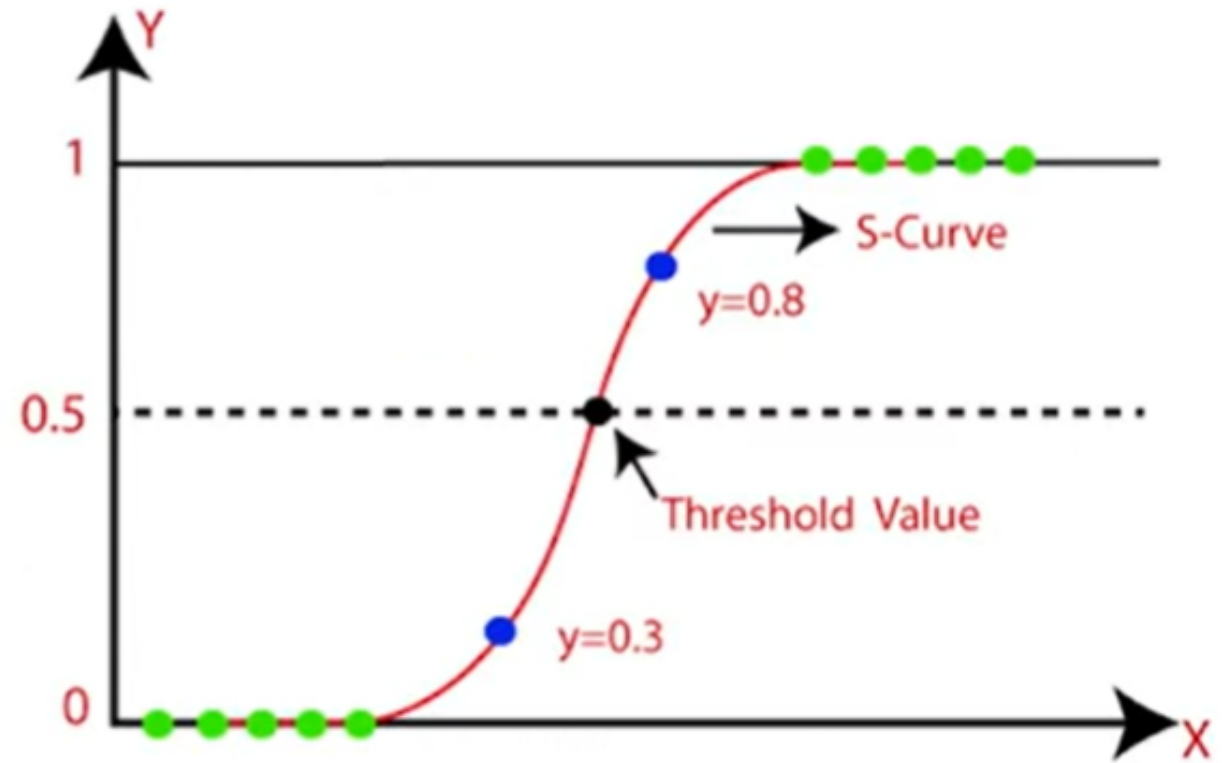
Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Assume the model suggested by the optimizer for odds of passing the course is,

$$\log(odds) = -64 + 2 * hours$$

- We use Sigmoid Function in logistic regression

- $s(x) = \frac{1}{1+e^{-x}}$



1. Calculate the probability of pass for the student who studied 33 hours.

$$\bullet \quad p = \frac{1}{1+e^{-z}} \qquad s(x) = \frac{1}{1+e^{-x}}$$

$$\bullet \quad z = -64 + 2 * 33 = -64 + 66$$

$$\bullet \quad p = \frac{1}{1+e^{-2}} = 0.88$$

- That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$\log(odds) = z = -64 + 2 * hours$$

2. At least how many hours student should study that makes he will pass the course with the probability of more than

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$
- $\ln(e^{-z}) = \ln(0.0526)$

$$\ln(e^x) = x$$

$$-z = \ln(0.0526) = -2.94$$

$$z = 2.94$$

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- $z = 2.94$
- $\log(\text{odds}) = z = -64 + 2 * \text{hours}$
- $2.94 = -64 + 2 * \text{hours}$
- $2 * \text{hours} = 2.94 + 64$
- $2 * \text{hours} = 66.94$
- $\text{hours} = \frac{66.94}{2}$
- ***hours = 33.47 Hours***

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability

THANK YOU !!